Ratios, Proportions & Percentages

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Simplifying ratios**

Simplify the ratio $45 : 60 : 75$.

GCD of $45, 60, 75$ is 15: $3 : 4 : 5$.

2 **[Foundational]** • **Target: Simplifying ratios**

Express "12 boys to 18 girls" as a ratio in simplest form.

$12 : 18$, and GCD is 6, so $2 : 3$.

3 **[Foundational]** • **Target: Sharing a quantity in a given ratio**

Share 150 in the ratio $2 : 3$.

$2 + 3 = 5$ parts, each part $= 150 \div 5 = 30$. Shares: 60 and 90.

4 **[Foundational]** • **Target: Sharing a quantity in a given ratio**

Share 220 among three people in the ratio $3 : 4 : 4$.

$3 + 4 + 4 = 11$ parts, each part $= 220 \div 11 = 20$. Shares: $60, 80, 80$.

5 **[Foundational]** • **Target: Sharing a quantity in a given ratio**

A prize of 900 riyals is shared between Sara and Layla in the ratio $5 : 4$. Find each share.

$5 + 4 = 9$ parts, each part $= 900 \div 9 = 100$. Sara receives 500 riyals; Layla receives 400 riyals.

6 **[Foundational]** • **Target: Sharing a quantity in a given ratio**

Divide 84 kg of flour in the ratio $1 : 2 : 4$.

$1 + 2 + 4 = 7$ parts, each part $= 84 \div 7 = 12$ kg. Shares: 12 kg, 24 kg, 48 kg.

7 **[Foundational]** • **Target: Equivalent ratios and proportions**

A map has scale $1 : 50{,}000$. A distance of 3 cm on the map represents how many meters in reality?

$3 \times 50{,}000 = 150{,}000$ cm $= 1{,}500$ m.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Converting between fractions, decimals and percentages

Write $\frac{3}{8}$ as a percent.

$\frac{3}{8} = 0.375 = 37.5\%$.

9 [Intermediate] • Target: Converting between fractions, decimals and percentages

Write 0.45 as a fraction in lowest terms.

$0.45 = \frac{45}{100} = \frac{9}{20}$.

10 [Intermediate] • Target: Converting between fractions, decimals and percentages

Write $\frac{7}{40}$ as a decimal.

$\frac{7}{40} = 0.175$.

11 [Intermediate] • Target: Percentage increase and decrease

A salary of 4,000 riyals increases by 6%. Find the new salary.

$4{,}000 \times 1.06 = 4{,}240$ riyals.

12 [Intermediate] • Target: Percentage increase and decrease

A population of $2{,}500$ decreases by 12%. Find the new population.

$2{,}500 \times 0.88 = 2{,}200$.

13 [Intermediate] • Target: Finding percentage change and original values

A price rises from 80 to 92. Find the percentage increase.

Change $= 92 - 80 = 12$. Percent increase $= \frac{12}{80} \times 100\% = 15\%$.

14 [Intermediate] • Target: Finding percentage change and original values

A quantity falls from 250 to 200. Find the percentage decrease.

Change $= 250 - 200 = 50$. Percent decrease $= \frac{50}{250} \times 100\% = 20\%$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Finding percentage change and original values • Cross-Lesson: 5.01 Quantitative Comparisons

After a 20% discount, a jacket costs 96 riyals. Find the original price.

Original $\times 0.8 = 96$, so original $= 96 \div 0.8 = 120$ riyals.

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16 [Challenging] • Target: Finding percentage change and original values • Cross-Lesson: 6.01 Sequences & Patterns

After a 15% increase, a value becomes 138. Find the original value.

Original $\times 1.15 = 138$, so original $= 138 \div 1.15 = 120$.

17 [Challenging] • Target: Successive percentage changes • Cross-Lesson: 7.01 Data Analysis & Probability

A price of 100 rises 10%, then falls 10%. Find the final price.

$100 \times 1.1 \times 0.9 = 99$ — one riyal less than the start, since the second change applies to a larger base.

18 [Challenging] • Target: Successive percentage changes • Cross-Lesson: 8.01 Word Problems & Reasoning

A stock worth 200 riyals rises 20% one year, then rises a further 25% the next year. Find the final value.

$200 \times 1.2 \times 1.25 = 300$ riyals.

19 [Challenging] • Target: Successive percentage changes • Cross-Lesson: 1.01 Arithmetic & Number Sense

A quantity decreases by 10%, then decreases by a further 10%. Find the overall percentage decrease from the original amount.

$100 \times 0.9 \times 0.9 = 81$, a fall of 19 from 100 — an overall decrease of 19% (not 20%).

20 [Challenging] • Target: Successive percentage changes • Cross-Lesson: 3.01 Algebra Essentials

A price of 400 riyals increases by 5%, then decreases by 5%. Find the final price, and state whether it equals the starting price.

$400 \times 1.05 \times 0.95 = 399$ riyals — one riyal less than the original 400, so it is not equal.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Simplifying & writing ratios

A recipe uses 250 g flour and 2 eggs. Write the ratio of flour (in grams) to eggs in simplest form.

a. $1:125$

b. $250:2$

c. $125:1$ ✓

d. $2:250$

$250:2$ simplifies by dividing both by 2: $125:1$.

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22 [Foundational] • Target: MCQ — Simplifying & writing ratios

Express $0.75 : 2$ as a ratio of whole numbers in simplest form.

a. $3 : 4$

b. $3 : 8$ ✓

c. $75 : 2$

d. $75 : 200$

Multiply both terms by 4 to clear the decimal: $3 : 8$.

23 [Foundational] • Target: MCQ — Sharing in a given ratio

The bar chart shows the ratio in which a total of 84 is to be shared. Find the larger share.

a. 36

b. 21

c. 48 ✓

d. 63

Total parts $= 3 + 4 = 7$; each part $= 84 \div 7 = 12$. Larger share $= 4 \times 12 = 48$.

24 [Foundational] • Target: MCQ — Sharing in a given ratio

Share 260 SAR in the ratio $2 : 3 : 5$. Find the smallest share.

a. 78 SAR

b. 52 **SAR** ✓

c. 130 SAR

d. 65 SAR

Total parts $= 2 + 3 + 5 = 10$; each part $= 260 \div 10 = 26$. Smallest share $= 2 \times 26 = 52$.

25 [Foundational] • Target: MCQ — Sharing in a given ratio

Two business partners share a profit of $18{,}000$ SAR in the ratio $5 : 7$. Find the larger share.

a. $10{,}500$ **SAR** ✓

b. $9{,}000$ SAR

c. $7{,}500$ SAR

d. $12{,}857$ SAR

Each part $= 18{,}000 \div 12 = 1{,}500$. Larger share $= 7 \times 1{,}500 = 10{,}500$ SAR.

26 [Foundational] • Target: MCQ — Sharing in a given ratio

A line of length 36 cm is divided in the ratio $4 : 5$. Find the length of the shorter piece.

a. 16 ✓

b. 4

c. 9

d. 20

Total parts $= 4 + 5 = 9$; each part $= 36 \div 9 = 4$ cm. Shorter piece $= 4 \times 4 = 16$ cm.

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27 [Foundational] • Target: MCQ — Sharing in a given ratio

A class of 40 students has boys and girls in the ratio 3 : 5. How many boys are there?

a. 25

b. 8

c. 24

d. 15 ✓

Total parts $= 3 + 5 = 8$; each part $= 40 \div 8 = 5$. Boys $= 3 \times 5 = 15$.

28 [Foundational] • Target: MCQ — Equivalent ratios & scale

A map has scale 1 : 50,000. A distance of 3 cm on the map represents how many km in reality?

a. 15 km

b. 1.5 **km** ✓

c. 150 km

d. 0.15 km

$3 \times 50{,}000 = 150{,}000$ cm $= 1{,}500$ m $= 1.5$ km.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Equivalent ratios & scale

Which ratio is equivalent to 4 : 9?

a. 9 : 4

b. 20 : 45 ✓

c. 8 : 16

d. 16 : 37

4 : 9 scaled by 5 gives 20 : 45.

30 [Intermediate] • Target: MCQ — Equivalent ratios & scale

In a photo, a model that is 1.8 m tall appears 6 cm high. Using the same scale, how tall (in m) is a building that appears 40 cm high in the photo?

a. 4.8

b. 24

c. 7.2

d. 12 ✓

Scale: 6 cm represents 1.8 m, so 1 cm represents 0.3 m. $40 \times 0.3 = 12$ m.

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31 [Intermediate] • Target: MCQ — Percent of a number

The bar chart shows four students' quiz scores out of 80. Which score is exactly 75%?

a. 68

b. 64

c. 60 ✓

d. 56

75% of 80 = 0.75 × 80 = 60.

32 [Intermediate] • Target: MCQ — Percent of a number

The bar chart shows monthly rent for four apartments. 8% of which apartment's rent equals 200 SAR?

a. Apt. 2

b. Apt. 1

c. **Apt. 3** ✓

d. Apt. 4

0.08 × 2,500 = 200 SAR, matching Apartment 3's rent of 2,500 SAR.

33 [Intermediate] • Target: MCQ — Percent of a number

150% of a number is 300. Find the number.

a. 250

b. 450

c. 200 ✓

d. 150

$1.5 \times x = 300 \Rightarrow x = 300 \div 1.5 = 200$.

34 [Intermediate] • Target: MCQ — Fraction, decimal & percent conversion

Write 85% as a fraction in simplest form.

a. $\frac{17}{20}$ ✓

b. $\frac{8}{5}$

c. $\frac{17}{25}$

d. $\frac{85}{100}$

$85\% = \frac{85}{100}$, which simplifies (divide by 5) to $\frac{17}{20}$.

35 [Intermediate] • Target: MCQ — Fraction, decimal & percent conversion

Which of these is largest: $\frac{3}{8}$, 0.4, or 37%?

a. 0.4 ✓

b. They are all equal

c. $\frac{3}{8}$

d. 37%

$\frac{3}{8} = 0.375 = 37.5\%$, and $37\% = 0.37$. So 0.4 is the largest.

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36 [Intermediate] • Target: MCQ — Percentage increase & decrease

The bar chart shows a salary before and after a raise. What percentage increase does it represent?

a. 15%

b. 500%

c. 20% ✓

d. 25%

Increase $= 6{,}000 - 5{,}000 = 1{,}000$. Percentage increase $= \frac{1{,}000}{5{,}000} \times 100\% = 20\%$.

37 [Intermediate] • Target: MCQ — Percentage increase & decrease

A shirt costs 120 SAR after a 20% discount. Find the original price.

a. 150 **SAR** ✓

b. 144 SAR

c. 100 SAR

d. 140 SAR

$120 = 0.8 \times \text{original} \Rightarrow \text{original} = 120 \div 0.8 = 150$ SAR.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Percentage increase & decrease • Cross-Lesson: 8.01 Word Problems & Reasoning

Find the percentage change from 40 to 34.

a. 17.6% decrease

b. 15% increase

c. 6% decrease

d. 15% **decrease** ✓

Change $= 34 - 40 = -6$. Percentage change $= \frac{-6}{40} \times 100\% = -15\%$, a 15% decrease.

39 [Challenging] • Target: MCQ — Finding the original value • Cross-Lesson: 1.01 Arithmetic & Number Sense

After a 15% increase, a price is 92 SAR. Find the original price.

a. 80 **SAR** ✓

b. 78.20 SAR

c. 88 SAR

d. 107 SAR

$92 = 1.15 \times \text{original} \Rightarrow \text{original} = 92 \div 1.15 = 80$ SAR.

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40 [Challenging] • Target: MCQ — Finding the original value • Cross-Lesson: 3.01 Algebra Essentials

After a 30% discount, an item costs 56 SAR. Find the original price.

a. 72.80 SAR

b. 80 **SAR** ✓

c. 86 SAR

d. 74.67 SAR

$56 = 0.7 \times \text{original} \Rightarrow \text{original} = 56 \div 0.7 = 80$ SAR.

41 [Challenging] • Target: MCQ — Finding the original value • Cross-Lesson: 4.01 Geometry & Measurement

A population grew by 8% to reach 27,000. Find the population before the growth.

a. 25,000 ✓

b. 26,000

c. 29,160

d. 24,840

$27{,}000 = 1.08 \times \text{original} \Rightarrow \text{original} = 27{,}000 \div 1.08 = 25{,}000$.

42 [Challenging] • Target: MCQ — Finding the original value • Cross-Lesson: 5.01 Quantitative Comparisons

After including a 15% service charge, a restaurant bill is 138 SAR. Find the bill before the charge.

a. 158.70 SAR

b. 120 **SAR** ✓

c. 123 SAR

d. 117.30 SAR

$138 = 1.15 \times \text{original} \Rightarrow \text{original} = 138 \div 1.15 = 120$ SAR.

43 [Challenging] • Target: MCQ — Successive percentage changes • Cross-Lesson: 6.01 Sequences & Patterns

A price of 200 rises 10%, then rises another 10%. Find the final price.

a. 220

b. 240

c. 244

d. 242 ✓

$200 \times 1.1 \times 1.1 = 242$.

44 [Challenging] • Target: MCQ — Successive percentage changes • Cross-Lesson: 7.01 Data Analysis & Probability

A price of 500 falls 20%, then rises 20%. Find the final price.

a. 460

b. 480 ✓

c. 500

d. 520

$500 \times 0.8 \times 1.2 = 480$ — a rise then fall of the same percentage never returns to the start.

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45 [Challenging] • Target: MCQ — Successive percentage changes • Cross-Lesson: 8.01 Word Problems & Reasoning

The bar chart shows a stock's value at the start and after rising 50% in year one. It then falls 50% in year two. Find its final value.

a. 500

b. 900

c. 1,000

d. 750 ✓

$1{,}000 \times 1.5 \times 0.5 = 750$.

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